4. Differentiate between elements and compounds.
5. Write any five characteristics of compounds.
6. Compare the properties of metals and non-metals. Give three examples for each.
7. Write down the properties of metalloids.

VII. Rewrite the given sentence in correct form.

1. Elements contain two or more kind of atoms and compounds contain only one kind of atom.

VIII. Higher Order Thinking Skills.

1. List out the metals, non-metals and metalloids which you use in your house, schools. Compare their properties.
2. What changes take place in the movement and arrangement of particles during heating process?
3. In the diagram given below, the circle, square and triangle represent the atoms of different elements.

Identify all combinations that represent
   a. molecule of a compound
   b. molecule of an element consisting of two atoms
   c. molecule of an element consisting of three atoms

4. Aakash noticed that the metal latch on gate was difficult to open during hot sunny days. However, it was not difficult to open the same latch at night. Aakash observed that the latch and the gate are exposed to the sun during day time.
   a. Formulate a hypothesis based on the information provided.
   b. Briefly state how you would test the hypothesis.

IX Consider the following statements and choose the correct option.

1. **Assertion:** Oxygen is a compound.
   Reason: Oxygen cannot be broken down into anything simpler.
2. **Assertion:** Hydrogen is an element.
   Reason: Hydrogen cannot be broken down into anything simpler.
3. **Assertion:** Air is a compound.
   Reason: Air consists of carbon dioxide.
4. **Assertion:** Air is a mixture of elements only.
   Reason: Only nitrogen, oxygen and neon gases exist in air.
5. **Assertion:** Mercury is solid in room temperature.
   Reason: Mercury is a non-metal.

   a. Both assertion and reason are true and reason is the correct explanation of assertion.
   b. Both assertion and reason are true, but reason is not the correct explanation of assertion.
   c. Assertion is true but reason is false.
   d. Assertion is false but reason is true.

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